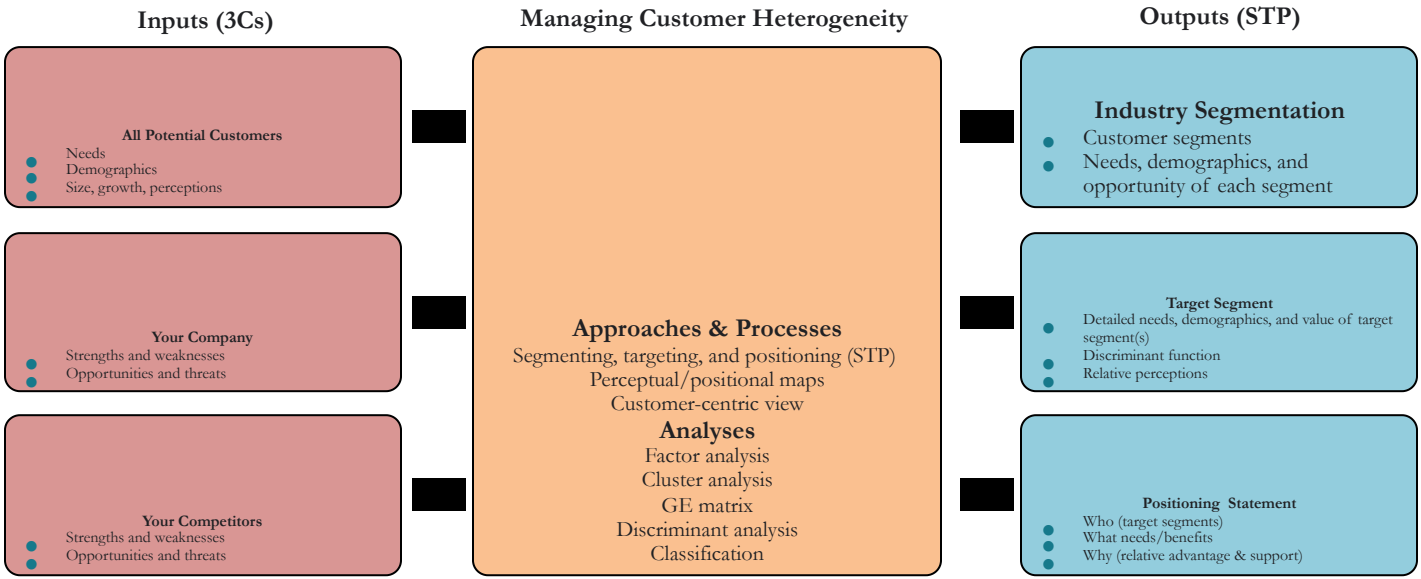
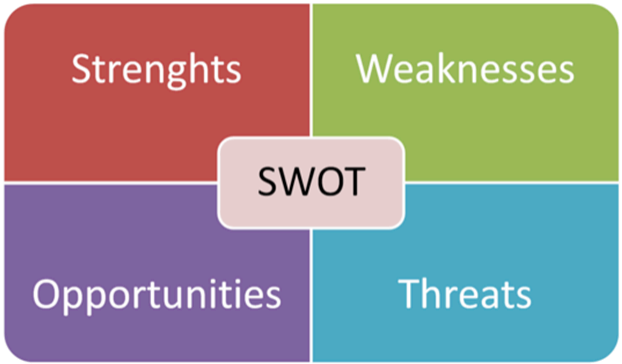
Marketing Principle #1: All Customers Differ → Managing Customer Heterogeneity



Inputs to the Managing Customer Heterogeneity Framework

- All potential **c**ustomers -- needs, desires, and preferences across customers in an industry, geographic region, market segment, or product category
- Your **c**ompany -- an inventory of the company's **s**trengths, **w**eaknesses, **o**pportunities, and **t**hreats (SWOT analysis)
- Your **c**ompetitors -- an inventory of your competitor's **s**trengths, **w**eaknesses, **o**pportunities, and **t**hreats (SWOT analysis)
-



SWOT and 3C Analyses

SWOT

Organizes internal strengths and weaknesses with external opportunities and threats that may affect a decision.

3C analysis

Examines the company, customers, and competitors to identify where the firm can create distinctive value.

Inputs	Internal capabilities and constraints	Customer needs and market change	Competitor strengths and likely responses

SWOT Analysis

Strengths	Weaknesses
Current internal advantages that the firm can use.	Current internal limitations that require improvement or mitigation.
Opportunities	**Threats**
External developments that may create value.	External developments that may reduce performance.

Translate each observation into an implication for the segmentation, targeting, or positioning decision.

Example: Bakery SWOT Analysis

A bakery is considering a new location in another neighborhood.

Strength Distinctive, well-known house-baked sourdough.	**Weakness** High production cost and limited parking at the proposed location.
Opportunity A nearby grocery store may bring new customer traffic.	**Threat** Established competitors already have loyal local customers.

Example: Bakery 3C Analysis

The same decision looks different when managers organize the evidence around the three market actors.

Company

Known for house-baked sourdough, but production costs are high.

Customers

Neighborhood families may value quality, convenience, and products for children.

Competitors

A donut shop and a café already offer convenient local alternatives.

Outputs of Managing Customer Heterogeneity Framework

- **Industry Segmentation** describes industry segments and includes, for each named segment, salient purchase preferences, demographic variables, and potential demand opportunities
  - → How can the marketplace be described using homogenous groups?
  - → What does each group of potential customers want?

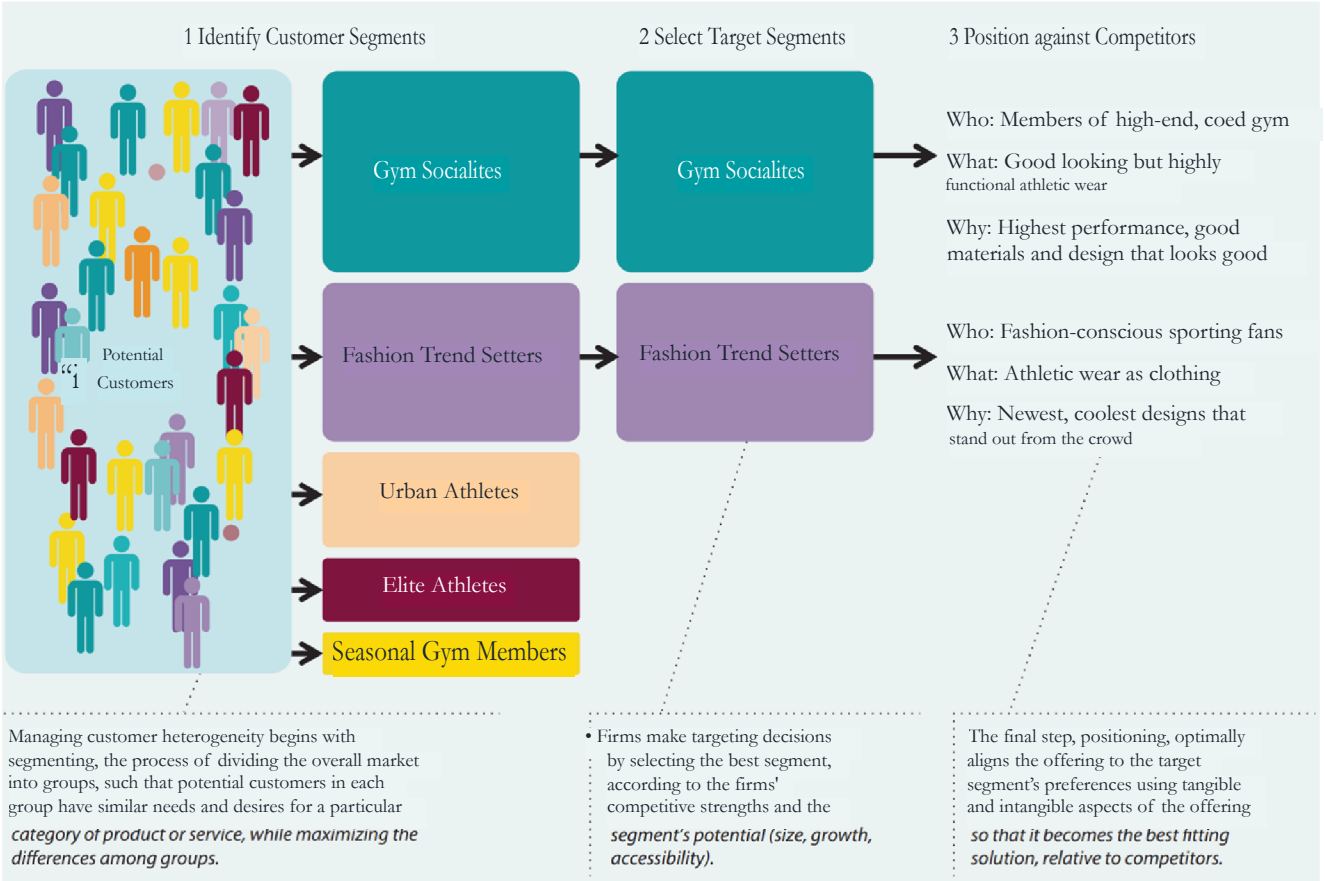
- **Target Segmentation** moves from the overall market landscape to the specific segment(s) of interest, such that it extends the first output by providing a very detailed description of each target segment.
  - → What set of segments will the firm pursue?
  - → How does the firm identify each group of target customers?

- **Positioning Statements** encapsulate the three questions into one concise statement that firms use to direct their internal and external marketing activities: who should the firm target, what needs and benefits are being fulfilled, and why does this offering provide a relative advantage over competitive offerings

Process for Managing Customer Heterogeneity

- To convert the inputs into outputs, marketers conduct a series of process steps
  - → **Segmenting** – To initiate the segmentation, managers need to *identify the key purchase attributes*, that is the needs and desires that a potential customer evaluates when making a purchase decision for this category
  - → **Targeting** – The targeting process follows naturally from segmentation, *to identify which segments the firm wants to sell to, based on the attractiveness of each segment and the firm's competitive strength in each segment*
  - → **Positioning** – The separation between targeting and positioning is often blurry. Many of the factors used to evaluate competitive strengths to select a target segment also impact the difficulty of executing an effective positioning strategy for that segment
  - → **Building Customer Centricity** – Building a customer-centric organization is different from executing an STP process, in that it requires a top-down, enduring commitment from senior leaders to institute a customer-centric philosophy across the firm's entire organization

Example of Managing Customer Heterogeneity



Agenda

01 **Framework**
Inputs, outputs, and process for managing customer heterogeneity

02 **AI Frontiers**
Embedding-based segmentation of unstructured customer data

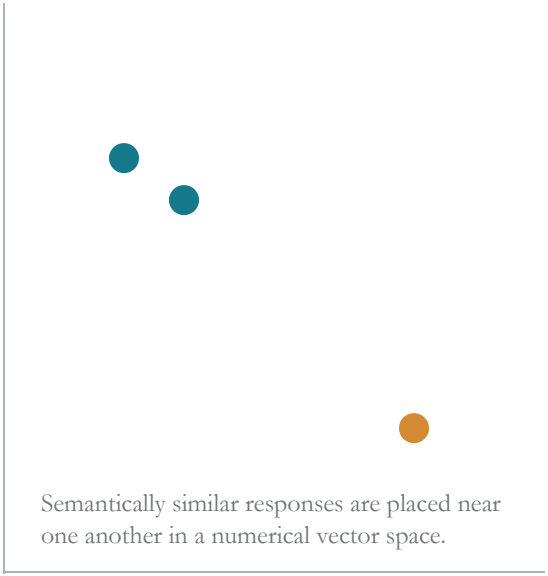
03 **Homework workshop**
Guided EMBA segmentation case from survey design to targeting

AI Frontiers: The Intuition Behind Embeddings

> "I need a flexible schedule because I travel every week."

> "Career mobility matters more than campus amenities."

> "I want a strong peer network in my industry."



Semantically similar responses are placed near one another in a numerical vector space.

Conceptual basis: Reimers and Gurevych, Sentence-BERT, EMNLP 2019.

Embedding-Based Segmentation Workflow

1. Collect

Open-ended responses, reviews, call transcripts, or product descriptions

→

2. Embed

Convert each customer's text into a dense semantic vector

→

3. Cluster

Use k-means, hierarchical clustering, or a density-based method

→

4. Interpret

Name, profile, and validate the segments against business outcomes

Captures meaning beyond exact keywords

Uses unstructured data at scale

Can complement survey variables

Example: Embedding MBA Applicants' Open-Ended Goals

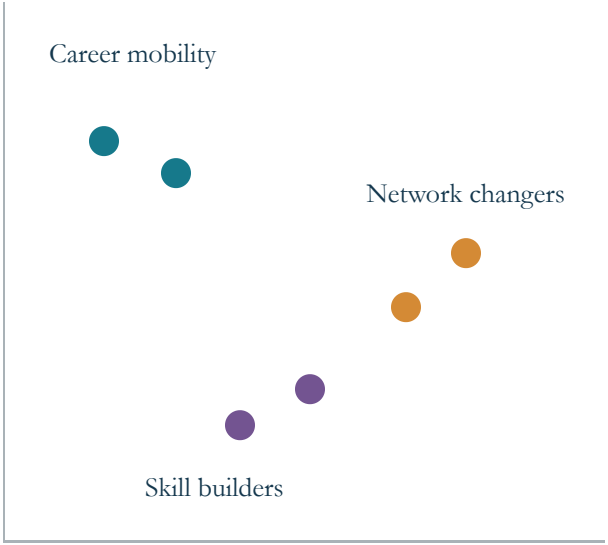
Applicant responses

"Move into management without leaving my job."

"Change industry and build a new network."

"Gain credibility for a promotion."

"Learn analytics for my current role."



> The clusters reveal language and motives that a fixed survey may not anticipate. Managers still need to test whether the groups differ in behavior, value, or response.

Limits and Validation of Embedding-Based Segmentation

Model dependence

Different embedding models and prompts may change the geometry and the clusters.

Interpretability

Cluster labels come from human judgment and can oversimplify mixed motives.

Bias and privacy

Text can reveal sensitive attributes and reproduce patterns from the training data.

Business validity

Semantic separation matters only if segments differ in size, value, needs, or response.

Validate stability across models and samples, inspect representative responses, and link clusters to observable outcomes.

Homework Workshop: From Survey Data to STP

Guided in class
Completed as homework

1. Check the survey data and missing values.
2. Use factor analysis to reduce related need ratings.
3. Use cluster analysis to identify applicant segments.
4. Profile segments with discriminant or classification analysis.
5. Select a target with the GE matrix.
6. Use a perceptual map to support positioning.

Homework Case: EMBA Enrollment

Business problem

You have been asked to develop a marketing program that increases enrollment in an Executive MBA program.

Your task

- Identify applicant segments from the survey.
- Choose a target segment using explicit criteria.
- Recommend a positioning direction supported by the data.

> We will complete the workflow together in class. Your homework documents the decisions and interpretation at each step.

Qualitative Questions: Probing for Important Attributes

general



specific

- Please discuss the process of deciding to get an Executive MBA.
- How did you begin your search for an Executive MBA program?
- What were important factors you considered when deciding on a specific EMBA program, and why were these factors important?
- What made you choose the Executive program over the full-time program?
- Of the schools you considered, what were the most compelling qualities of top choices?

Segmentation Questions

- Key attributes when making a decision (needs and benefits)
- All potential customers report on importance of attributes

Please rate your agreement with the following statements:	**Strongly Disagree**						**Strongly Agree**	***construct***
A low-cost MBA program is very important to me.	1	2	3	4	5	6	7	*cost*
It is very important that an MBA program have a rigorous curriculum.	1	2	3	4	5	6	7	*rigor*
An easily accessible MBA program is important to me.	1	2	3	4	5	6	7	*location*
I think it is very important that an MBA program fits with my schedule.	1	2	3	4	5	6	7	*convenience*
It is very important that an MBA program is prestigious.	1	2	3	4	5	6	7	*prestige*
I feel it is very important that I am challenged by the coursework of an MBA program.	1	2	3	4	5	6	7	*rigor*
It is important that an MBA program is not expensive.	1	2	3	4	5	6	7	*cost*
I think it is very important that an MBA program be highly respected.	1	2	3	4	5	6	7	*prestige*

Discriminant/Classification Questions

- Key demographics questions

What is your age?	_____
Please indicate your gender:	__male __female
What is your annual salary, including bonuses and commissions in US dollars?	__$0 - $25,000 __$25,001 - $50,000 __$50,001 - $75,000 __$75,001 - $100,000 __$100,001 - $125,000
How many miles is your home from the UW campus?	_____
How many years of work experience do you have?	
What was your undergraduate major?	_____

Data Needed For Target Market Selection

- Segment Attractiveness
  - → Growth rate of segments
  - → Price sensitivity of segment
- Competitive Strength
  - → UW relative advantage
  - → Fit to UW brand image
- Where does this data come from?
  - → Survey
  - → Secondary sources
  - → SWOT and 3Cs

Perceptions and Preference Questions for Positioning Maps

- Across key attributes what do they think of you and the competitors and preference
- Can do on all potential customers or just target customers

Please evaluate each University's EMBA program (repeat for UW, Berkeley, etc):

Measure perceptions…

	Strongly disagree						Strongly Agree
Program is low cost	1	2	3	4	5	6	7
This is a high rigor program	1	2	3	4	5	6	7
The location is very good	1	2	3	4	5	6	7
Program is very convenient	1	2	3	4	5	6	7
Program is very prestigious	1	2	3	4	5	6	7

Measure preferences…

	low						high
Likelihood of selecting	1	2	3	4	5	6	7

Segmentation: Create a Picture of Each Cluster

Segmentation variable / Cluster	Overall	Cluster 1	Cluster 2	Cluster 3
Low cost	4.8	6.67	1.21	6.79
High rigor	3.28	1.33	6.93	1.29
Good location	3.2	1.33	1.21	6.79
Very convenient	2.9	6.75	1.21	1.29
High prestige	5.2	1.33	6.93	6.79

Discriminant variable / Cluster	Overall	Cluster 1	Cluster 2	Cluster 3
Age	29.1	23.667	35.857	27
Gender	1.325	1.167	1.286	1.5
Salary	3.9	2.833	5.214	3.5
Miles from Campus	7.7	3.25	13.214	6
Work Experience	6.85	2.083	12.286	5.5

Cluster 1 ("young professionals") is concerned with cost and convenience and is younger with lower income and less job experience.
Cluster 2 ("returning executives") is primarily concerned with the rigor and prestige of the EMBA program. These customers tend to be older with higher salaries and more work experience.
Cluster 3 ("busy manager") wants a prestigious program with a close location at a reasonable price. They've got a little more experience and are making a reasonable salary.

Targeting Using GE Matrix

Horizontal Axis



Competitive Strength	Young Professional	Returning Executive	Busy Manager
Fit to UW strengths	1.5	4.5	3
Fit to UW brand image	1.5	4	4

Vertical Axis



Segment Attractiveness	Young Professional	Returning Executive	Busy Manager
Growth rate of segment	2	4	3
High margins	2	4.5	1

Bubble Size



	Young Professional	Returning Executive	Busy Manager
Market Size	5	20	10

Which Segment Should You Target?

